

CARBOHYDRATES

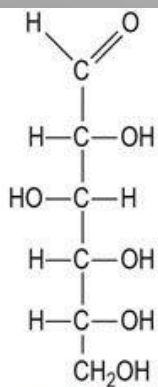
COLLEGE OF MEDICINE - HMU

1ST YEAR MEDICAL STUDENTS

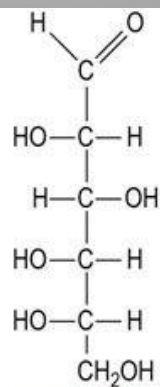
PREPARED BY

SHELAN I. SABER

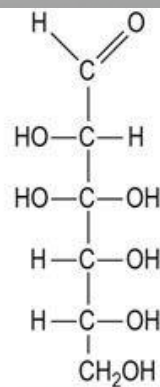
MSC. IN BIOCHEMISTRY



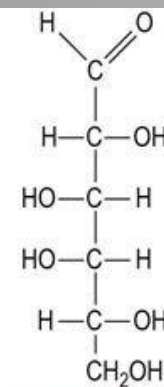
D-Glucose



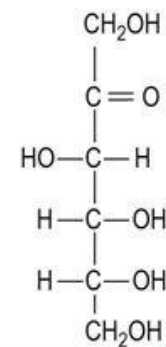
L-Glucose



D-Mannose



D-Galactose



D-Fructose

CARBOHYDRATES

Definition:

- Carbohydrates are compounds containing carbon, hydrogen, and oxygen with a general empirical formula of: $C_n(H_2O)_n$.
- Carbohydrates are polyhydroxy aldehydes or polyhydroxy ketones or compounds which produce them on hydrolysis.
- Carbohydrates are a major source of energy from our diet.

CARBOHYDRATES ARE CLASSIFIED AS FOLLOW:-

1- Monosaccharides:-

Are the simplest forms of sugar they maybe subdivided into triose, tetrose, pentose, hexoses and heptose depending upon the number of carbon atoms they possess as aldoses or ketoses.

Examples of monosaccharides include:
glucose, fructose, and galactose.

2- Disaccharides

Are disaccharide carbohydrates composed of two monosaccharides residue united by glycosidic linkage. Some important disaccharides are:-

Sucrose = glucose + fructose

cane or beet sugar

Maltose = glucose + glucose

malt sugar

Lactose = glucose + galactose

Milk sugar

3- Oligo saccharide:-

Carbohydrates that contain between 3 and 10 monosaccharides.

Example: Raffinose

4- Polysaccharides:-

are very large polymers composed of more than 10 molecules of monosaccharides joined together by glycosidic linkages.

Example:

Starch, glycogen, dextrose, cellulose, etc....

Medical importance of carbohydrates:-

Glucose is a simple sugar that serves as a primary fuel for energy production, especially for the brain, muscles and several other body organs and tissues. Glucose also serves as a building block for larger structural molecules of the body, such as glycoproteins and glycolipids.

carbohydrates having high specific function such as:-

- Glycogen for storage

- Ribose in nucleic acid

- Galactose in lactose of milk, in certain complex lipids and in combination with protein in glycoprotein.

Diseases associated with carbohydrates include:-

- Diabetes mellitus
 - Galactosemia
 - Glucose-6-phosphate dehydrogenase deficiency (favism)
 - Fructosuria
 - glycogen storage diseases
- 

Qualitative tests for carbohydrates

Test no. 1

Molisch test / general test for carbohydrates

Principle:-

Concentrated H_2SO_4 hydrolyze glycoside bonds to give the monosaccharide and dehydrated to furfural or its derivatives (hydroxyl methyl furfural), this product combined with α -naphthol to give a reddish violet zone, appears at the junction between the two liquids.

Procedure

1. Add 3 ml of sugar solution
2. Add 2 drops of alpha-naphthol
3. Incline the solution and add 2 ml of Conc. H_2SO_4 to flow down the side of the tube.

Thus forming a violet ring between the two liquids.



Test no. 2

Iodine test (to distinguish between mono-saccharide and poly-saccharide)

Principle:-

Iodine form characteristic colored complex with poly saccharides. Starch gives blue, glycogen brownish red and dextrin purple or red color with iodine solution, while simple sugars do not change the color of iodine solution. These colors disappear on heating and reappear on cooling.

Procedure

1. Add 2-3 ml of starch solution
2. Add 1 drop of diluted iodine solution
3. Observe the production the color changes.

Iodine Test: Principle, Procedure, Results

Brown/Yellow



Test Negative/ Starch Absent

Blue/Purple



Test Positive/ Starch Present

Test no. 3

Benedict's test (is used as a test for the presence of reducing sugars)

Principle:-

The copper sulfate (CuSO_4) present in benedict's solution reacts with electrons from the aldehyde or ketone group of the reducing sugar in alkaline medium to form insoluble red copper oxide (Cu_2O).

Procedure

1. Place 2 ml of benedict's reagent.
2. Add exactly 4 drops of sugar solution.
3. Heat the tube in water bath for 3 min.
4. Allow the fluid to cool.
- 5 After cooling the color maybe red yellow or orange depending upon the amount of sugar present.



Test no. 4

Barfoed's test (to distinguish between monosaccharide and disaccharide)

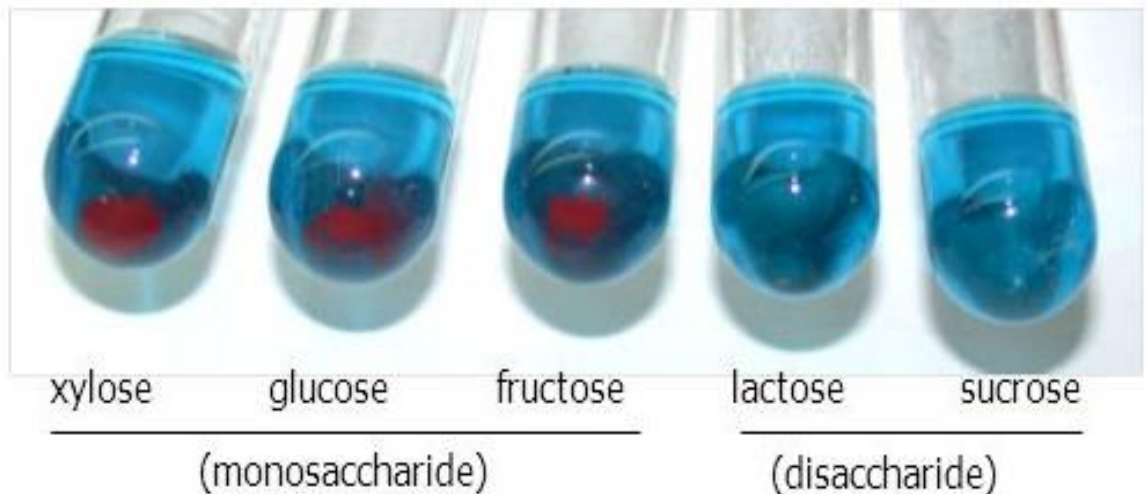
Principle:-

This test serving to detect monosaccharide. The same redox reaction (benedict's test) but in weak acid medium. Disaccharide will also respond to the test under proper condition of acidity and boiling the sugar solution more than needed which is enough to hydrolyze the disaccharide to give the positive test.

Procedure

1. Place 2 ml of barfoed's reagent.
2. Add 4 drops of sugar solution.
3. Heat the tube in water bath for 2 min.
4. Reduction is indicated by production of a red precipitate of Cu_2O .

Barfoed's test
(test for
monosaccharides)



Test no. 5

Bails test (for pentose)

Principle:-

When pentose are heated with concentrated HCL, furfural is formed which condenses with orcinol in the presence of (Fe^{+3}) ions to give a green color.

Procedure

1. Place 2 ml of bails solution
2. Add 1 ml of sugar solution
3. Heat the tube in water bath for 2 min.
4. Immediately on cooling, the solution becomes green in the presence of pentose sugar.



Test no. 6

Seliwanoff's test (for ketohexoses)

Principle:-

Ketohexoses (fructose) are converted to hydroxymethyl furfural in hot hydrochloric acid, which couples with resorcinol to produce a red color compound.

Procedure

1. Place 2 ml of seliwanoff's solution.
2. Add 2 drops of sugar solution and heat the solution in water bath for 2 min.
3. A positive reaction is indicated by the production of a red color with or without brown-red precipitate.

